Intermediate Elective

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Set Up

Packages

```r
# general use
library(tidyverse)
library(snakecase)
library(janitor)

# file organization
library(here)

# reading .xlsx files
library(readxl)

# data visualization
library(naniar)
library(patchwork)
library(gghighlight)
library(paletteer)
library(ggplot2)
```

Data

```r
# taxonomic info
taxon_list <- read_csv(here("data","taxon_list.csv"))
```

```
# aquatic invertebrates
aq_ins <- read_xlsx(
  here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "Aquatic Insects")

# site info
sites <- read_xlsx(
  here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "sites")
```

Problems

1. Explain your inspiration

I was inspired by Tobias Stalder's "Hiking Locations in Washington" because the figure's characteristics would allow me to show average abundance/L data for a specific aquatic invertebrate species, separated by NCOS location, while also highlighting a mean average abundance/L value for each site.

This figure makes sense for the aquatic invertebrate data set because it allows for a clear comparison of average abundance/L between different sites, just as the original figure is separated by hiking locations.

The plot will show average abundance/L of ostracods across NCOS locations, the x axis will be NCOS location and the y axis will be average abundance/L. Each jitter point will represent a different average abundance/L measurement, and the mean average abundance/L for each location will be highlighted with a different color to stand out.

2. Sketching Figure

ViewPDF

3. Code your figure

Cleaning

```r
taxon_list_clean <- taxon_list |>
  # convert taxon names to snakecase
  mutate(verbatim_name = to_any_case(verbatim_name,
  case = "snake"))

# creating new object
ncos_sites <- sites |>
# filtering to only include ncos quarterly sampling sites
filter(sector == "NCOS" & sampling_frequency == "Quarterly")

# water quality
water_quality <- read_xlsx(
  here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "Water Quality",
  na = c("", "n/a", "N/A", "over", "173+"))

water_quality_clean <- water_quality |>
  clean_names() |>
  semi_join(ncos_sites, by = "site") |>
  unite("date_site", date, site, remove = FALSE) |>
  group_by(date, site, date_site) |>
  summarize(med_ph = median(p_h, na.rm = TRUE),
            med_sal = median(salinity_ppt, na.rm = TRUE),
            med_do = median(dissolved_oxygen_mg_l, na.rm = TRUE)) |>
  ungroup() |>
  filter(date_site != "2022-08-12_NVBR")

# creating new clean object from aquatic inverts
aq_ins_clean <- aq_ins |>
  # clean column names
  clean_names() |>
  # filtering join: only include sites occurring in ncos_sites
  semi_join(ncos_sites, by = c("site")) |>
  # renaming columns
  rename(date = date_on_vial) |>
  # select columns of interest
  select(date, sample_type, site,
        ostracod,
        copepod,
        hemiptera_corixidae_boatman,
        cladocera,
        nematode,
        diptera_ceratopogonidae,
```

```r
         annelida_oligochaete,
         diptera_chironomid,
         annelida_polychaete) |>
  # filter to only include "filtered beaker" samples
  filter(sample_type %in% c("FB 250", "FB250")) |>
  # convert the data frame to long format
  pivot_longer(cols = ostracod:annelida_polychaete,
               names_to = "taxon",
               values_to = "abundance") |>
  # group by date, site, taxon
  group_by(date, site, taxon) |>
  # calculate average abundance per liter (sum abundance divided by 7.5
  # liters)
  summarize(ave_lit = sum(abundance, na.rm = TRUE)/7.5) |>
  # ungroup data frame
  ungroup() |>
  # create a new column called `date_site` to join with water quality
  unite("date_site", date, site, remove = FALSE) |>
  # join with water quality
  left_join(
  select(water_quality_clean,
         date_site,
         med_ph,
         med_sal,
         med_do),
         by = "date_site") |>
  # join with taxon list
  left_join(taxon_list_clean, by = c("taxon" = "verbatim_name")) |>
  # add full names of sites
  mutate(site_full = case_when(
         site == "NVBR" ~ "North of Venoco Bridge",
         site == "NEC" ~ "South of Venoco Bridge",
         site == "NMC" ~ "Main Channel",
         site == "NPB" ~ "South of Phelps Bridge",
         site == "NPB1" ~ "North of Phelps Bridge",
         site == "NPB2" ~ "Phelps Road",
         site == "NWP" ~ "West Pond",
         site == "NDC" ~ "Devereux Creek")) |>
  # setting factor levels for site
  mutate(site_full = fct_reorder(site_full, med_sal, .fun = "median", .na_rm
= TRUE),
  site_full = fct_rev(site_full))
```

Wrangling

```
# create new object from aq_ins_clean
ost_abundance <- aq_ins_clean |>
  # filter to only ostracods
  filter(taxon == "ostracod") |>
  # remove missing abundance/L values
  filter(!is.na(ave_lit)) |>
  # group by site
  group_by(site_full) |>
  # log + 1 transform mean abundance per liter
  mutate(log_ave_lit = log(ave_lit + 1)) |>
  # ungroup data frame
  ungroup()
```

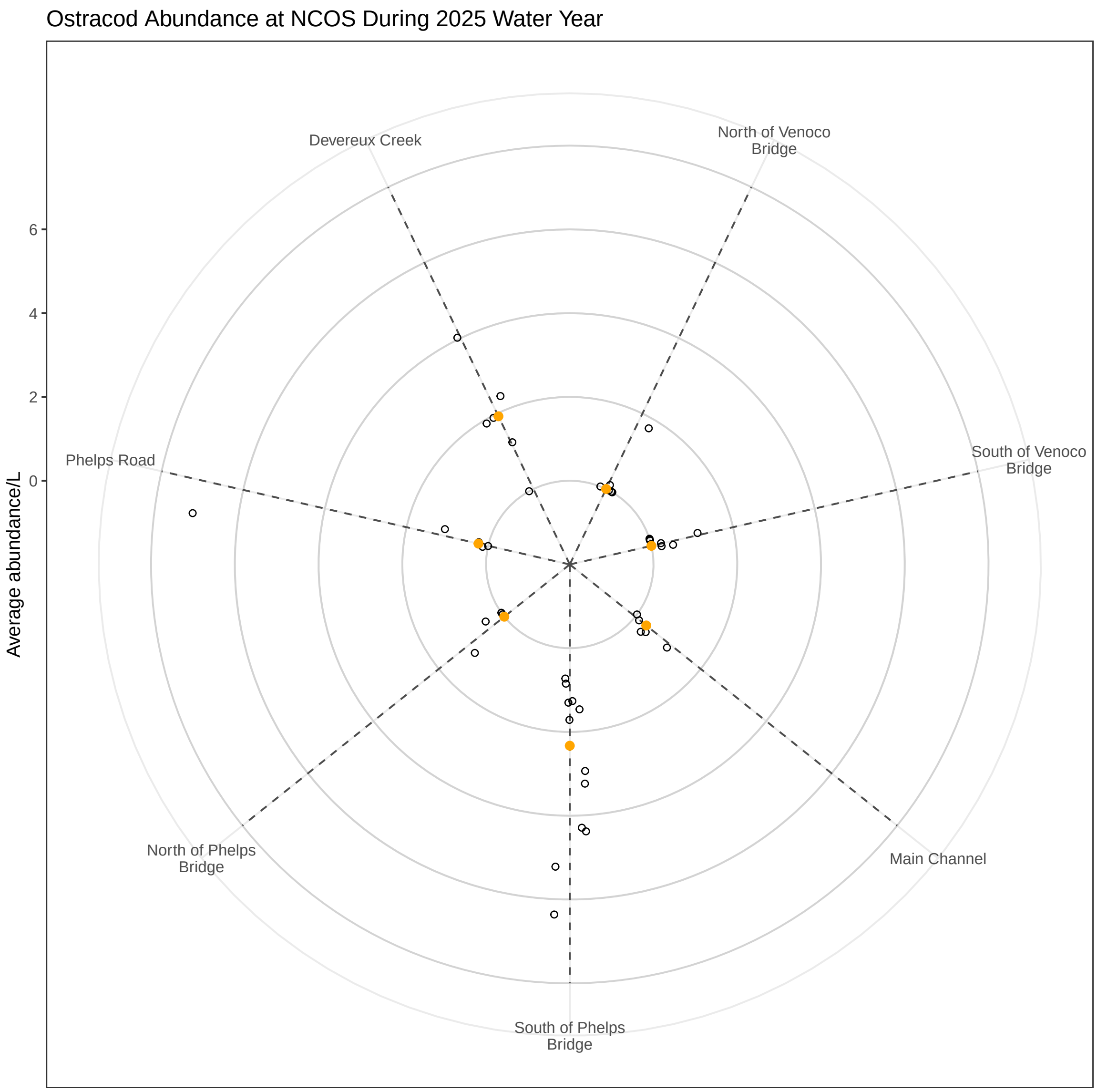
Plotting

```
# setting limits for ring plot, max values
y_max <- max(ost_abundance$log_ave_lit, na.rm = TRUE)
ring_log <- pretty(c(0, y_max), n = 4)

# creating new plot
ostracods_plot <- ggplot(
  # inserting ostracod data frame
  data = ost_abundance,
  # setting x axis as site and y axis as average abundance/L
  mapping = aes(
    x = site_full,
    y = log_ave_lit)) +
  # setting ring geometry, color
   geom_hline(
    yintercept = ring_log,
    color = "lightgray") +
  # setting geometry for line for each site
  geom_vline(
    xintercept = 1:length(unique(ost_abundance$site_full)),
    linetype = "dashed",
    color = "gray30") +
  # setting geometry for jitter data points, sixe and shape
  geom_jitter(
```

```r
    height = 0,
    width = 0.1,
    shape = 21) +
  # creating mean average abundance/L data point in different color
  stat_summary(
    geom = "point",
    fun = "median",
    size = 2,
    color = "orange") +
  # displaying site names on x axis
  scale_x_discrete(labels = \(x) str_wrap(x, 15)) +
  #scale y axis so bars don't start in the center
  scale_y_continuous(limits = c(-2, 8),
                     expand = c(0,0),
                     breaks = c(0, 2, 4, 6))+
  # labeling axes and setting title
  labs(x = "",
    y = "Average abundance/L",
    title = "Ostracod Abundance at NCOS During 2025 Water Year") +
  # turning plot into a circle (polar coordinates)
  coord_polar() +
  # changing theme
  theme_bw()

#displaying plot
ostracods_plot
```



4. Describe your process

In 1-3 sentences each, describe:

- your process of using someone else's code to create a visualization
- challenges you encountered as you made your visualization
- how your visualization differs from visualizations you made in class, for assignments, or for a previous elective